

SAMPLING AND LABORATORY PROTOCOL OF BENTHONIC FAUNA OF INVERTEBRATES IN FORDABLE RIVERS

CODE: ML-Rv-I-2013

Approved by instruction of the Secretary of State for the Environment dated November 22, 2013



**GOBIERNO
DE ESPAÑA**

**MINISTERIO
DE AGRICULTURA, ALIMENTACIÓN
Y MEDIO AMBIENTE**

This document is part of a series of sampling, laboratory and index and metric calculation protocols for use in monitoring programs for the status of continental water bodies (rivers, lakes and reservoirs) and in the classification of their ecological status.

The specifications of these documents must be taken into account by the River Basin Organizations in the operation of the official networks for monitoring the state and ecological potential in the continental surface water bodies, either directly or through service contracts.
These protocols are subject to changes deemed necessary in light of scientific progress in the field.



MINISTRY OF AGRICULTURE, FOOD AND ENVIRONMENT

Edited

by: © Ministry of Agriculture, Food and Environment
Technical General Secretariat
Publications Center

Catalogue of Publications of the General State Administration: <http://publicacionesoficiales.boe.es/>

NIPO: 770-11-310-8



INDEX

1. APPLICABILITY	5
2. OBJECTIVE	5
3. REFERENCE REGULATIONS	5
4. EQUIPMENT, REAGENTS AND PRESERVATIVES	6
4.1. FIELD WORK	6
4.2. LABORATORY WORK	7
5. SELECTION AND DELIMITATION OF THE SAMPLING POINT.....	7
6. SAMPLING FREQUENCY AND TIME	8
7. SAMPLING PROCEDURE	8
7.1. CHARACTERIZATION OF THE SAMPLING AREA AND IDENTIFICATION OF HABITAT TYPES	8
7.2. SAMPLING.....	9
8. STORAGE, LABELLING AND TRANSPORTATION OF SAMPLES	10
9. LABORATORY SAMPLE ANALYSIS PROCEDURE	11
9.1. SAMPLE WASHING AND SIEVING	11
9.2. IDENTIFICATION AND COUNTING OF TAXONS	11
10. DATA PROCESSING	12
ANNEX I: FIELD SHEET FOR SAMPLING	13
ANNEX II: FIELD SHEET FOR IN SITU TAXONOMIC IDENTIFICATION	17
ANNEX III: LABORATORY SHEET	21



1. APPLICABILITY

This sampling and laboratory protocol is mandatory in the operation of official networks for the evaluation of ecological status/potential in compliance with Directive 2000/60/EC, the Water Framework Directive, which are operated by the Hydrographic Confederations (CCHH), either directly or through service contracts.

The stations where this protocol will be used are those belonging to the surveillance control program, operational control program, research control program and reference networks.

This protocol corresponds to the sampling and laboratory analysis of benthic invertebrate fauna in water bodies in the river category, as well as to artificial or highly modified water bodies similar to rivers that are fordable, and is applicable to obtaining samples for the classification of ecological status or ecological potential.

The sampling of this protocol is aimed at obtaining data on the composition and abundance of benthic macroinvertebrates, which are the group used in the classification of ecological status. These are invertebrates of a relatively large size (visible to the human eye), no less than 0.5 mm. They mainly include arthropods (insects, arachnids and crustaceans) together with oligochaetes, hirudineans and molluscs and, less frequently, coelenterates, bryozoans or platyhelminthes.

The information collected through this protocol provides valid data for calculating the metrics established in the Hydrological Planning Instruction (Order 2656/2008) for the quality element corresponding to the composition and abundance of benthic invertebrate fauna:

Iberian Biological Monitoring Working Party (IBMWP-2013).

Multimeter type specific (METI).

This sampling protocol can also be applied to calculate other benthic invertebrate metrics that require composition and abundance data such as:

Multimetric IMMi-L (qualitative) and IMMi-T (quantitative).

Multimetric devices used in the Mediterranean GIG (ICM-9, ICM-Star, ICM-7).

Iberian Average Score Per Taxon (IASPT). Margalef

Diversity Index (MARGALEF).

2. OBJECTIVE

Directive 2000/60/EC, the Water Framework Directive, requires Member States to implement monitoring programmes. These programmes must enable the composition and abundance of benthic invertebrate fauna to be monitored and assessed.

The Water Framework Directive establishes that the methods used to control the parameters of each type must comply with international or national standards that guarantee the provision of information of equivalent quality and scientific comparability.

Therefore, the objective of this protocol is to establish a method of sampling benthic invertebrates in rivers that guarantees compliance with the requirements mentioned above.

3. REFERENCE REGULATIONS

The reference regulations of this protocol are those listed below:



Directive 2000/60/EC of the European Parliament and of the Council establishing a framework for Community action in the field of water policy.

Legislative Royal Decree 1/2001 approving the Revised Text of the Water Law.

RD 907/2007 approving the Hydrological Planning Regulations.

Order MAM/3207/2006 approving ITC-MMA EECC-1/06 Complementary technical instruction on chemical and microbiological determinations for water analysis.

Order ARM/2656/2008 approving the Hydrological Planning Instruction.

Order MAM/985/2006, which develops the legal regime of the collaborating entities of the hydraulic administration in matters of control and surveillance of water quality and management of discharges into the public hydraulic domain.

This protocol has been drafted taking into account the following standards:

UNE – EN 10870: 2012 – Guidelines for the selection of methods and devices for sampling benthic macroinvertebrates in freshwater.

UNE – EN 5667-1: 2007 – Part 1. Guide for the design of sampling programmes and sampling techniques.

UNE – EN 14996: 2007 – Guide for quality assurance of biological and ecological assessments in the aquatic environment.

UNE – EN 8689-1: 2000 – Part 1. Guide for the interpretation of data relating to biological quality from studies of benthic macroinvertebrates.

UNE – EN 8689-1: 2000 – Part 2. Guide for the presentation of data relating to biological quality from studies of benthic macroinvertebrates.

UNE – EN 16150:2012 – Guidelines for sampling benthic macroinvertebrates in wadeable rivers by apportioning the cover surfaces of the habitats present.

4. EQUIPMENT, REAGENTS AND PRESERVATIVES

4.1. FIELD WORK

Equipment and materials for sample collection

Macroinvertebrate sampling net with a 500 µm mesh size, frame with a base of 0.25 m and a height equal to or greater than the net and long handle. 2 PVC trays (minimum 30 x 20 cm).

Entomological tweezers.

1-2 L wide-mouth, watertight containers for storing macroinvertebrate samples.

Wash bottle.

Formaldehyde (HCHO) 40% / Ethyl alcohol (C₂H₅OH) 96%.

Sodium borate Na₂B₄O₇ x (10 H₂O).

Multiparameter probe with temperature, conductivity, pH and dissolved oxygen sensors.

Sampling protocol.

Field leaves.

Identification keys for biological quality elements (MAGRAMA).

Equipment and complementary material

Ballpoint pen, permanent marker, or other method for labeling samples. If labels are used, they must be moisture-resistant.

Waterproof covers for field tokens.

GPS.



Digital camera.
Adequate cartography.
Mobile phone.
Scissors.
Adhesive tape and onion skin paper to label the samples.
Containers suitable for transporting sample bottles and fixative.
Wader.
Long latex and rubber gloves (above the elbow).

All material used in the field must be properly cleaned and disinfected to prevent the transport and dispersion of propagules or individuals of invasive species, following the protocols established by the competent river basin authority.

4.2. LABORATORY WORK

Equipment for sample analysis

Sieves of 5 mm, 1 mm and 0.5 mm.
3 PVC trays (minimum 30 x 20 cm).
Elutriator or hydroseparator.
Subsampler (Wrona et al., 1982) optional.
Ethyl alcohol (C₂H₅OH) at 96% and 70%.
Formaldehyde (HCHO) 40%.
Sodium borate Na₂B₄O₇ x (10 H₂O).
50 ml pipette.
Petri dishes.
Entomological tweezers.
Plastic vials and other containers with tight-fitting stoppers.
Identification keys for biological quality elements (MAGRAMA).
Laboratory sheets.
Graduated rack.

Equipment and complementary material

Respiratory protection mask with filters for organic compounds (if formaldehyde is used to preserve samples).

Safety glasses.
Latex gloves.
Fume hood or aspirator with filters for organic compounds in the sample washing area.

Permanent markers.
Adhesive tape.
Scissors.

For both field and laboratory work, all necessary measures must be taken to ensure that work is carried out under appropriate safety and hygiene conditions.

5. SELECTION AND DELIMITATION OF THE SAMPLING POINT

The sampling point will be a selected 100 m stretch representative of the characteristics of the water body.



The selected section will present the most frequent habitat types in the water body, so that it is representative of the natural variability of physical and structural elements (for example the rapid-pool sequence, etc.). The following aspects will be taken into account:

The sampling section will reflect the dominant fast-slow sequence.

The river morphology and habitat composition will be the characteristics of the section to be evaluated, for example, channeled areas will be avoided if the rest of the section is not.

The vegetation cover (density, shade) will be characteristic of the section; this will prevent sampling a shaded area if this is not common in the section.

Areas immediately adjacent to bridges, fords or weirs should be avoided, unless they are characteristic of the section. If possible, samples should be taken upstream of the access point.

Accessible sampling points will be sampled. Sampling will be carried out in wadeable areas, taking the necessary precautions and avoiding risks.

The selected section will be delimited by recording the UTM coordinates (measured with GPS) of the starting point.

6. SAMPLING FREQUENCY AND TIME

Sampling will be carried out annually in spring, when macroinvertebrate communities usually reach their maximum diversity. Exceptionally, sampling may be postponed until early summer to find a more favourable situation in cases where meteorological or hydrological conditions so require (mainly in mountain areas with high rainfall or snow influence).

In the case of a recent flood, at least 15 days must be allowed before taking samples. In temporary rivers, it is very important to adapt the sampling time to suitable hydrological conditions that guarantee the existence of a continuous water flow.

7. SAMPLING PROCEDURE

The sampling and habitat characterization procedure is described below.

7.1. CHARACTERIZATION OF THE SAMPLING AREA AND IDENTIFICATION OF HABITAT TYPES

The sampling point will be visited and observations will be made of the presence of river habitats, as well as the characteristics of the banks. This visit will be carried out along the bank whenever it is accessible to avoid trampling the section before sampling.

A photographic report will be carried out and the field sheet included in Annex I of this protocol will be filled out.

Identification of habitat types: The identification of the types of habitat present in the section will be carried out taking into account the following five groups:

Hard substrates (): rocks, stones and gravel predominate in areas of rapids, characteristic of most mountain and foothill riverbeds. Dominant in most upper courses and less common in lower courses.

Plant debris (leaf litter, logs of different sizes) (): Detritus and other plant remains that have remained submerged for a relatively long time (not recently fallen) provide excellent colonization.

Vegetated banks (): submerged banks, with roots and emergent plants associated with them.



Submerged macrophytes (*): are seasonal and may not be present in all channels, particularly in the upper reaches.

Sand and other fine sediments (): generally in low current areas and associated with the banks, although it may be predominant in some channels.

Once the types of habitat and the area occupied by each of them have been identified, the sampling units (kicks) to be carried out will be distributed among the different habitats present in the section. As a general rule, twenty sampling units will be carried out.

The distribution of the sampling units in the 5 types of habitats will be carried out in proportion to the area occupied by each one in the sampling station, so that each sampling unit corresponds to 5% of the coverage area of a habitat.

Habitats with a representation greater than 5% are dominant and those less than 5% are minority. In the case of minority habitats, the adjustment will be made by combining fractions of sampling units (eg 0.5 for a habitat represented by a coverage between 2-3%).

In addition, the distribution of sampling units will take into account other factors that may influence habitat heterogeneity, such as water velocity and depth.

In each sampling unit, the substrate located half a metre in front of the mouth of the net, which has a base of 0.25 m, is removed. The final sampled area resulting from The twenty sampling units will be approximately 2.5 m².

7.2. SAMPLING

Sampling is carried out by going upstream (from downstream to upstream) and taking into account the number of sampling units and the distribution in the previously defined habitat types.

Before starting sampling, it is necessary to identify the macroinvertebrates that live on the surface of the water, or those that, even though they live submerged, are difficult to capture.

Each habitat type is sampled differently, as detailed below:

Hard substrates: gravel, pebbles and blocks are sampled by holding the lower edge of the net against the ground and dislodging organisms by stirring with feet or hands in shallow areas, a length of 0.5 m of substrate upstream of the net.

Plant debris (leaves, logs of different sizes): they are sampled by removing the debris deposits with feet or hands, keeping the net downstream (with current) or passing the net over them (in slow waters) to collect the suspended organisms.

Wood accumulated in pools is also sampled in this habitat, avoiding large pieces because they are generally difficult to sample properly.

Vegetated banks: river banks with roots and emergent plants associated with them. The roots are stirred with feet or hands within 0.5 m and any organisms suspended or carried by the current are collected with the net placed downstream.

Submerged macrophytes: these are sampled by dragging the net through the vegetation from the bed (where it is rooted) to the water surface (maximum 0.5 m). In shallow waters, sampling is carried out by shaking the plants with your feet or hands over a length of 0.5 m, and collecting any organisms in suspension or carried by the current with the net. Avoid resuspension of the sediment.

Sand and other fine sediments: The net is initially passed across the surface of the water to collect macroinvertebrates from the water surface. Deposition areas of non-vegetated sediment are sampled and agitated by foot or hand to include suspended material in the net, over a length of 0.5 m. Dragging the net through soft sediments should be avoided to reduce the amount of debris in the samples.



It should be noted that in waters with a current the speed of the water will drag the resuspended macroinvertebrates towards the inside of the net, but in the case of waters with little current the net will have to be moved manually along the 0.5 m of its path to capture the macroinvertebrates.

As the sampling units are completed, the net must be emptied into one or more trays to avoid losing organisms when taking new catches. The net must be emptied before the new sampling, especially in sections with little current and some depth.

The sample is observed and stones and large pieces of debris are carefully removed, avoiding at all times the loss of invertebrates from the sample.

Recording of sampling data Sampling data

must be recorded on the field sheet in Annex I. Information must be collected on:

Habitats present and distribution of sampling units: hard substrates, plant debris, vegetated banks, submerged macrophytes and sand and other fine sediments. The following must be specified: whether the waters are with strong or slow current, the percentage of occupation and the number of sampling units carried out (taking into account that they may include mixed habitats).

Field identification of captured taxa (visu) for which it will be necessary to use the field sheet in Annex II.

Invertebrates observed on the surface or submerged, when they have not been captured, for which it will be necessary to use the field sheet in Annex II.

Number of jars and sample code.

If there are discrepancies between the planned efforts and those that have been sampled (non-fordable habitats, subdivision of habitats, etc.) they must be reflected in the field sheet.

8. CONSERVATION, LABELLING AND TRANSPORTATION OF SAMPLES

Once the sampling is finished, the sample is placed in jars with a hermetic seal and wide mouth.

As a preservative you can use:

96% ethyl alcohol to be added to the sample filtrate once the excess water has been removed until a concentration of 70% v/v is obtained.

40% formaldehyde to be added to the sample with water until obtaining a concentration in the sample of 4% v/v. It is recommended to add only a few drops first to anesthetize the invertebrates and prevent them from adopting rigid postures that may make their identification difficult, and after a few minutes add the rest of the reagent. Formaldehyde is toxic and its use requires the application of safety measures. In the field, work will be done outdoors, with latex gloves, spills will be avoided and suitable airtight containers will be used. It is recommended to add sodium borate to the formaldehyde to prevent the calcareous parts of the organisms from being destroyed.

The jars are marked with two labels, one made of onionskin paper written in pencil on the inside and another on the outside written in indelible ink. Both labels, at least, must show: the sampling campaign code, the sample code, the date and, if more than one jar has been used to store the samples, this information must also be recorded.

When transporting samples from the field to the laboratory, the necessary measures will be taken to prevent breakage of the sample containers or the release of vapors. It is recommended to use airtight containers and store them in refrigerators or boxes with lids in a cool place, avoiding prolonged exposure to the sun.



9. LABORATORY SAMPLE ANALYSIS PROCEDURE

It is recommended that the time between sample collection and analysis not be excessive to avoid degradation of macroinvertebrates. As a general rule, the period between sample collection and analysis should never exceed one year.

If 96% ethyl alcohol is used, this period should be even shorter, especially if the sample has a high organic matter content.

Sample processing will be carried out following the steps below:

9.1. WASHING AND SIEVING OF SAMPLES

Using the necessary protective measures, the container is opened and the sample is poured onto the tower of sieves (5 mm, 1 mm and 0.5 mm), leaving the one with the least light at the bottom; the water with formaldehyde is collected for later treatment as toxic waste. Then, under the tap, the sample is washed with plenty of water, separating the organisms from the remains of debris, stones and sand that may have remained, and until the smell of formaldehyde disappears, and the sample has been separated into the respective fractions on the sieves.

The contents of each sieve are poured into the respective coarse (> 5 mm), medium (between 5 and > 1 mm) and fine (between 1 and 0.5 mm) fraction trays. To do this, the contents are washed to one side of the sieve before placing it in the tray and then washed, turning it over. The laboratory sheet is prepared, identifying the sample according to the label.

9.2. IDENTIFICATION AND COUNTING OF TAXONS

Coarse fraction: All individuals in the coarse fraction are extracted, identified and counted. The extracted material is stored in a vial with 70% ethyl alcohol, with the reference to the sampling point, date and indicating fraction G. In some cases (only when there is an abundance of algae or macrophytes) a sub-sampling of the coarse fraction can be carried out (once washed in the sieves). This is homogenised in a tray and subdivided into 2 or 4 fractions, of which 1 or 2 are inspected, ensuring a minimum count of 100 individuals, and subsequently inspecting the other fractions to see if there are taxa not counted in the analysed sub-sample. In this case, Gx will be indicated, where 1/x is the analysed portion of the coarse fraction.

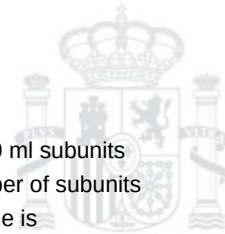
Middle fraction: If the fraction contains a lot of sand or gravel, the inorganic material (sand or gravel) is first separated from the organic material by means of elutriation. The middle fraction is then subsampled; for this, the sample is homogenised and divided into equal parts. A portion of the sample is then extracted (e.g. 1/8), ensuring that the taxa in the fraction are in the same proportion as in the entire sample and that it contains at least 100 invertebrates (following Wrona et al., 1982). If the number does not reach 100, a second or subsequent whole subsamples must be taken until at least 100 is reached. The abundance of each taxon is quantified.

The rest of the sample is then placed in a large tray and new taxa are extracted to complete the taxonomic list. The analyzed fraction is preserved in ethyl alcohol labeled M a, where 1/a is the analyzed portion of the total mean fraction¹.

Fine fraction: This is the most homogeneous fraction. To perform subsampling, the inorganic material (grit) must first be separated from the organic material by means of elutriation.

The inorganic substrates will then have to be checked for any remaining invertebrates (taking small portions with a Pasteur pipette and placing them on a Petri dish to examine them with a magnifying glass). Then the part containing the macroinvertebrates is homogenized by placing them in a liter of water, using the subsampler described above.

¹ If the fraction analyzed represents 1/8 of the total sample, a=8



by Wrona et al. (1982) or another comparable method. Once the sample has been homogenized, 2 or 3 50 ml subunits (each representing 1/20 of the total) are extracted using a 50 ml pipette and placed in Petri dishes. A number of subunits should be subsampled to obtain a number of individuals equal to or greater than 100, so that the subsample is representative of the sample (Wrona et al., 1982).

An alternative method of processing the fine fraction consists of carrying out the elutriation and testing of the inorganic substrates as described above, and proceeding to filter the suspended organic material through a 0.5 mm mesh sieve. It is convenient to have sieves of different diameters in order to use the one that best suits the amount of filtrate. The homogenization of the filtrate in the sieve is ensured by placing it on a tray (with a rack drawn on the bottom) with a little water and making circular movements to homogenize the sample. A fraction containing at least 100 individuals is then extracted. To do this, a tool must be used to fractionate the sample and extract the subsample with the help of a spatula and tweezers, washing the filter and sieve walls appropriately.

Next, it is necessary to count and identify the invertebrates of the subunits analysed (Wrona method) or of the observed fraction (alternative method). In both cases, the counts of each taxon must be recorded on the laboratory sheet and the fraction that the subsample represents of the total sample must be indicated.

The invertebrates of the fine fraction are stored in a vial with 70% alcohol, labeled F·b, where 1/b is the analyzed portion of the total fine fraction²

Identification of taxa will be carried out by observing morphological characteristics, using a binocular magnifying glass and following appropriate identification guides at the required level. The Key for the Identification of Biological Quality Elements prepared by the General Directorate of Water (ID-TAX) will be used as the main reference.

Once the sample has been treated in the laboratory, the results will be filled out on the laboratory sheet included in Annex III of this protocol.

10. DATA PROCESSING

The results of sampling and laboratory analysis will consist of:

Inventory of taxa obtained and their total abundance expressed as number of individuals of each taxon. The abundances obtained from samples analysed according to fractions will be calculated as the sum of the individuals of the three fractions multiplied by their dividends ($G + Ma + Fb$; or $Gx + Ma + Fb$).

Inventory of taxa (presence/absence) identified in the field (visu).

Inventory of identified taxa and their abundance in each of the fractions analyzed: coarse, medium and fine.

Completed field sheets and laboratory sheet.

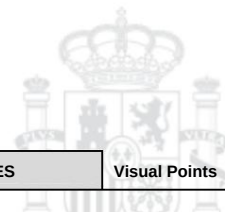
² If the fraction analyzed represents 1/20 of the total sample, $b=20$

ANNEX I: FIELD SHEET FOR SAMPLING



		MINISTRY AGRICULTURE, FOOD AND ENVIRONMENT		DATA COLLECTION SAMPLING: BENTHIC INVERTEBRATES IN FORDABLE RIVERS	
SAMPLING IDENTIFICATION DATA					
TYPE OF WATER BODY:				WATER BODY CODE:	
NAME OF THE BODY OF WATER:					
SAMPLING POINT CODE: X/Y COORDINATES (ETRS 89): /					SPINDLE:
ORGANIZATION/COMPANY:					
SAMPLER:			Program:	Surveillance:	
SAMPLE CODE:		NUMBER OF BOATS:		Operational:	
DATE: / /		Start time: :		Investigation:	
		End time: :		Reference:	
SAMPLE CONSERVATION: ☐ Ethyl alcohol ☐ Formaldehyde					
Access description and location of the section:					
PHYSICOCHEMICAL CHARACTERISTICS					
pH (units):			Dissolved oxygen (mg O ₂ /l):		
Water temperature (°C):			% O ₂ Saturation:		
Electrical conductivity at 20°C (µS/cm):					
Observations:					
HYDROMORPHOLOGICAL CHARACTERISTICS					
Average width (m) of the section:		Average depth (m) of the section:		Length (m) of the section:	
HABITATS	% COVERAGE	NUMBER OF UNITS OF SAMPLING		PHOTO CODE	
Hard substrates ()					
Plant debris (y)					
Vegetated shores ()					
Submerged macrophytes (*)					
Sand and other sediments ()					
PREDOMINANT WATER SPEED (mark with X)					
Null: Absence of flow					
Reduced: Laminar flow without ripples					
Moderate: Small symmetrical surface undulation					
Fast: Broken surface undulation					
Very fast: Fast, foaming					
Habitat comments:					

ANNEX II: FIELD SHEET FOR IDENTIFICATION IN SITU TAXONOMICS



CODE	ARACHNIDS	Points visual	CODE	EPHEMEROPTERANS	Points visual	CODE	ODONATES	Visual Points
ACA001SPOR	Acariformes ¹	4	BAE001FAMI	Baetidae	4	AES001FAMI	Aeschnidae	8
			CAE001FAMI	Caenidae	4	CAL004FAMI	Calopterygidae	8
			H002FAMI	Ephemerelellidae	7	COE001FAMI	Coenagrionidae	6
CHR009FAMI	Chrysomelidae	4	EPH001FAMI	Ephemeridae	10	COR012FAMI	Cordulegasteridae	8
CUR001FAMI	Curculionidae	4	HEP001FAMI	Heptageniidae	10	COR008FAMI	Corduliidae	8
DRY001FAMI	Dryopidae	5	LEP003FAMI	Lepthlebiidae	10	GOM003FAMI	Gomphidae	8
DYT001FAMI	Dytiscidae	3	OLI002FAMI	Oligoneuriidae	5	LES001FAMI	Lestidae	8
ELM001FAMI	Elmidae	5	POL020FAMI	Polymitarcidae	5	LIB001FAMI	Libellulidae	8
GYR001FAMI	Gyrinidae	3	POT003FAMI	Potamanthidae	10	PLA004FAMI	Platycnemididae	6
HAL002FAMI	Halplidae	4	PRO010FAMI	Prosopistomatidae	7			
HEL002FAMI	Helophoridae	5	SIP001FAMI	Siphonuridae	10	CODE	OLIGOCHAETES	Visual Points
HYD008FAMI	Hydraenidae	5					All	1
HYD013FAMI	Hydrochidae	5	CODE	HETEROPTERANS	Visual Points			
HYD011FAMI	Hydrophilidae	3	APH001FAMI	Aphelocheiridae	10	CODE	PLECOPTERA	Visual Points
HYG001FAMI	Hygrobiidae	3	COR004FAMI	Corixidae	3	CAP003FAMI	Capniidae	10
NOT004FAMI	Noteridae	3	GER002FAMI	Gerridae	3	CHL004FAMI	Chloroperlidae	10
PSE004FAMI	Psephenidae	3	HYD014FAMI	Hydrometridae	3	LEU004FAMI	Leuctridae	10
SCI001FAMI	Scirtidae (=Helodidae)	3	MES001FAMI	Mesoveliidae	3	NEM001FAMI	Nemouridae	7
			NAU001FAMI	Naucoridae	3	PER004FAMI	Peridae	10
						PER006FAMI	Perodidae	10
CODE	CRUSTACEANS	Visual punctuation	NEP002FAMI	Nepidae	3	TAE001FAMI	Taeniopterygidae	10
ASE001FAMI	Asellidae	3	NOT003FAMI	Notonectidae	3			
AST003FAMI	Astacidae	8	PLE004FAMI	Pleidae	3			
ATY001FAMI	Atyidae	6	VEL001FAMI	Velidae	3	CODE	TRICHOPTERA	Visual Points
COR003FAMI	Corophiidae	6				BER001FAMI	Beraeidae	10
GAM001FAMI	Gammaridae	6	CODE	HIRUDINEANS	Punt. visu BR	RA006FAMI	Brachycentridae	10
OST001CLAS	Ostracoda	3	ERP001FAMI	Erpobdellidae	3	CAL002FAMI	Calamoceratidae	10
PAL004FAMI	Palaeomonidae	6	GLO005FAMI	Glossophoniidae	3	ECN001FAMI	Ecnomidae	7
			HIR002FAMI	Hirudidae (=Hirudinidae) 3		GLO004FAMI	Glossosomatidae	8
			PIS003FAMI	Piscicolidae	4	GOE001FAMI	Goeridae	10
CODE	DIPTERA	Punt. visu				HYD006FAMI	Hydropsychidae	Punt. 5
ANT004FAMI	Anthomyiidae ²	4	CODE	NEUROPTERANS	visu	HYD012FAMI	Hydroptilidae	6
ATH001FAMI	Athericidae	10	SIA001FAMI	Sialidae	4	LEP008FAMI	Lepidostomatidae	10
BLE001FAMI	Blephariceridae	10				LEP004FAMI	Lepoceridae	10
CER006FAMI	Ceratopogonidae	4	CODE	LEPIDOPTERA	Punt. visu	LIM002FAMI	Limnephilidae	7
CHI001FAMI	Chironomidae	2	PYR004FAMI	Crambidae (=Pylalidae)	4	MOL001FAMI	Molannidae	10
CUL001FAMI	Culicidae	2				ODO001FAMI	Odonotoceridae	10
DIX001FAMI	Dixidae	4	CODE	MOLLUSCS	Punt. visu	PH001FAMI	Philopotamidae	8
DOL001FAMI	Dolichopodidae	4	ANC001FAMI	Ancylidae	6	PHR002FAMI	Phryganeidae	10
EMP001FAMI	Empididae	4	BIT001FAMI	Bithyniidae	3	POL003FAMI	Polycentropodidae	7
EPH003FAMI	Ephydriidae	2	FER002GENE	Ferrissia ³	6	PSY002FAMI	Psychomyiidae	8
LIM005FAMI	Limoniidae	4	HYD005FAMI	Hydrobiidae	3	RHY001FAMI	Rhyacophilidae	7
PSY001FAMI	Psychodidae	4	LYM001FAMI	Lymnaeidae	3	SER001FAMI	Sericoxomatidae	10
PTY001FAMI	Ptychopteridae	4	NER001FAMI	Neritidae	6	UEN001FAMI	Uenoidae (=Thremmatidae) 10	
RHA004FAMI	Rhagionidae	4	PHY003FAMI	Physidae	3			
SCA002FAMI	Scatophagidae ²	4	PLA003FAMI	Planorbidae ⁴	3	CODE	TOURBELARIANS	Visual Points
SCI002FAMI	Sciomyzidae	4	SPH006FAMI	Sphaeriidae	3	DEN001FAMI	Dendrocoelidae	5
SIM002FAMI	Simuliidae	5	THI001FAMI	Thiaridae	6	DUG001FAMI	Dugesidae	5
STR003FAMI	Stratiomyidae	4	UNI001FAMI	Unibidae	6	PLA005FAMI	Planariidae	5
SYR002FAMI	Syrphidae	1	VAL001FAMI	Valvatidae	3			
TAB002FAMI	Tabanidae	4	VIV001FAMI	Viviparidae	6			
THA003FAMI	Thaumaleidae	2						
TIP001FAMI	Tipulidae	5						

- The superorder Hydracarina has become the superorder Acariformes
- Anthomyiidae and Scatophagidae were previously grouped as Muscidae
- The Family Ferrissidae has become the Genus Ferrissia
- All genera except Ferrissia

MINISTRY
AGRICULTURE, FOOD
AND ENVIRONMENT

CODE	OTHERS	visual
	Nematode	
	Cnidaria - Hydridae	
	American crab	
	Crab signal	

[illegible]

ANNEX III: LABORATORY SHEET

DATA, PROCESSING AND SAMPLE IDENTIFICATION						
CODE		Taxon	Abundance			Total Abundance (**)
SISTCODSUP	SISTCODINF		G x= 1/ ____ (*)	M a= 1/ ____ (*)	F b= 1/ ____ (*)	

(*)Indicate the fraction analyzed. For example, x= 1/8 if it represents one eighth.

(**) The abundances of each fraction will be taken into account

OBSERVATIONS